JOHANNES KEPLER
UNIVERSITY LINZ

Submitted by
David Nadrchal

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Thesis Supervisor
Dipl.-Ing. Florian Schmid

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Automatic Speech Recognition for a Tracheostomy Patient: A Case Study

Bachelor Thesis

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JOHANNES KEPLER UNIVERSITY LINZ
Altenbergerstraße 69
4040 Linz, Austria
www.jku.at
DVR 0093696

Abstract

This bachelor thesis presents a case study on the development of a personalised *automatic speech recognition* (ASR) system for a Czech patient with a severe speech impairment caused by a permanent tracheostomy and potential neurological damage following a debilitating car accident. As we verify experimentally, the impairment prevents the patient from speaking intelligibly to an untrained Czech listener. We collect and curate a public[1] dataset containing twenty-four annotated hours of their speech. To enable efficient data acquisition, we introduce a novel data collection method called *artificial conversations*. We fine-tune the Whisper deep-learning automatic speech recognition model to recognise speech in this dataset, leveraging pretraining on both regular Czech speech and quasi-tracheostomy speech. Furthermore, we employ knowledge distillation from a masked language model to teach Whisper a more nuanced understanding of the Czech language, thus achieving near-healthy-voice accuracy for data collected under standard conditions (i.e., *artificial conversations*), and improving performance of the publicly available smallest Whisper models for regular Czech speech. We also discuss limitations in the real-world applicability of our system and propose several strategies to overcome them. We experiment with word-by-word transcription strategies that constrain the output space to existing Czech vocabulary as well as with strategies for transcribing entire sentences. We achieve performance that is considered helpful by the patient.

[1] https://zenodo.org/records/15225595

Kurzfassung

Diese Bachelorarbeit präsentiert eine Fallstudie zur Entwicklung eines personalisierten *Automatic Speech Recognition* (ASR)-Systems für eine tschechische Patientin mit einer schweren Sprachstörung, verursacht durch eine permanente Tracheotomie und mögliche neurologische Schäden infolge eines schweren Autounfalls. Wie wir experimentell nachweisen, verhindert die Störung, dass die Patientin für eine ungeschulte tschechischsprachige Person verständlich spricht. Wir erheben und kuratieren ein öffentliches[2] Datenset mit 24 annotierten Stunden ihrer Sprache. Zur effizienten Datenerhebung führen wir eine neuartige Methode ein, die wir *artificial conversations* nennen. Wir passen das Deep-Learning-Modell der automatischen Spracherkennung Whisper an, um Sprache in diesem Datenset zu erkennen, und nutzen dabei ein Pretraining sowohl auf regulärer tschechischer Sprache als auch auf quasi-tracheotomiebedingter Sprache. Darüber hinaus wenden wir Knowledge Distillation aus einem Masked Language Model, um Whisper ein nuancierteres Verständnis der tschechischen Sprache beizubringen. Auf diese Weise erreichen wir eine Genauigkeit nahe derjenigen gesunder Stimmen für unter Standardbedingungen (d. h. *artificial conversations*) erhobene Daten. Gleichzeitig verbessern wir die Genauigkeit der öffentlich verfügbaren kleinsten Whisper-Modelle für reguläre tschechische Sprache. Wir diskutieren zudem Einschränkungen bei der praktischen Anwendbarkeit unseres Systems und schlagen mehrere Strategien vor, um diese zu überwinden. Wir experimentieren sowohl mit Wort-für-Wort-Transkriptionsstrategien, die den Ausgaberaum auf existierenden tschechischen Wortschatz beschränken, als auch mit Strategien zur Transkription ganzer Sätze. Wir erreichen eine Genauigkeit, die von der Patientin als hilfreich gefunden wird.

[2] https://zenodo.org/records/15225595

Contents

Abbrevations and Notation

AAC	Alternative and Augmentative Communication
ASR	Automatic Speech Recognition
CER	Character Error Rate
CNN	Convolutional Neural Network
CTC	Connectionist Temporal Classification
DL	Deep Learning
DNN	Deep Neural Network
HMM	Hidden Markov Model
KD	Knowledge Distillation
LLM	Large Language Model
LSTM	Long Short-Term Memory
ML	Machine Learning
MLM	Masked Language Model
TSNE	t-Distributed Stochastic Neighbor Embedding
PCA	Principal Component Analysis
RNN	Recurrent Neural Network
SNR	Signal-to-Noise Ratio
WER	Word Error Rate

1. Introduction

This thesis is a case study of a Czech teenage girl ('Tali') who was struck by a car at a preschool age. The accident left her quadriplegic, and she requires a permanent tracheostomy to help her breathing. The adverse effect of the tracheostomy is, however, a severe speech impediment which is, possibly, further amplified by the impaired motorics resulting from her spine injuries. Although the family relatives or experienced assistants can learn to understand her speech, for a general Czech speaker, Tali's speech is unintelligible. We discuss her condition in more detail in Chapter 2.

Figure 1.1.: Face-swapped images of Tali. The picture on the left highlights the stoma on her throat. The picture on the right depicts her communication with experienced assistants.

In this thesis, we describe the development of a deep learning automatic speech recognition (ASR) system that would help Tali to verbally communicate even with untrained conversational partners. For this purpose, we organise over one hundred speaking sessions to collect 24 hours of annotated recordings of her speech. Afterwards, we fine-tune Whisper [1], an ASR [2] transformer by OpenAI, on this data. For smaller Whisper models, we achieve near-healthy-voice accuracy for the data collected during the standard recording sessions (i.e., artificial conversations detailed in Section 4.1.2).

We also discuss the results of our system on real-world conversations and propose several strategies how tackle the challenges caused by the domain shift. With these strategies, we achieve transcription accuracy that is already considered useful by Tali and her caretakers. Namely, we achieve 46% top-5 word accuracy in word-by-word transcription.

All our codes are available on GitHub[1]. Furthermore, we publish the speech collected from our patient on Zenodo [2].

1.1. Contribution

The key contributions of this work are:

- Creation and curation of a public dataset of Tali's speech that can also be used by other researchers and assistive technology developers.
- Demonstrating that knowledge distillation (KD) [3] from masked language models (MLM) [4] into Whisper can improve the ASR quality, especially when the Whisper model is relatively small. As a part of this contribution, we improve Whisper Tiny for regular Czech speech by 6% word error rate (WER). We provide our improved model to the general public via HuggingFace [3].
- Achieving near healthy-voice ASR performance on the in-domain data (i.e., artificial conversations) with Whisper Tiny and Whisper Base.
- Discussing challenges that come with the deployment of the system in the real world and providing and testing strategies on how to handle them.

[1] https://github.com/Hobit2002/TracheoSpeech_ASR
[2] https://zenodo.org/records/15225595
[3] https://huggingface.co/Hobit2002/whisper_tiny_cs

1.2. Outline

The rest of this thesis is structured as follows: In Chapter 2, we provide more detail on the causes of Tali's speech impediments. We also organised a small experiment to estimate the intelligibility of her speech to a general Czech speaker. In Chapter 3, we explore ASR methods while giving special focus to state-of-the-art DL models Deep Speech [5], Wave2Vec [6] and Whisper [1]. Furthermore, we investigate the contemporary assistive tools for people with severe speech impediments, ranging from classical and widely used low-tech tools to recent and more experimental methods. In Chapter 4, we describe in detail the data collection process, especially the *artificial conversations*, the predominant type of our recording sessions. In Chapter 5, we analyse Tali's speech, both quantitatively and qualitatively, and compare it to the standard one. In Chapter 6, we provide details on how we train our models, namely the pre-training steps and the KD from MLM. In Chapter 7, we explain our data split and provide the results of different models for each of regular Czech speech, quasi-tracheostomy speech and the speech of our patient. In Chapter 8, we discuss the remaining challenges in deploying the model and present the decoding strategies with which we achieve practically applicable results. In Chapter 9, we summarise our results and provide possible directions for future work.

2. Tali's condition

We briefly summarize the available information on Tali's condition[1] as it appears to be very specific and significantly deviating from standard descriptions of tracheostomy and other common speech impediments.

Tracheostomy is a surgical procedure performed to assist individuals with severe breathing difficulties, such as mechanically obstructed upper airways or respiratory failure. As we do not have access to Tali's medical records, the reason she required the procedure remains unknown. During the surgery, a stoma is created in the patient's trachea, and a plastic tube is inserted into it. Typically, the tracheostomy is maintained for only a few days or weeks, and the stoma closes naturally several days after the tube is removed [8].

Figure 2.1.: Tracheostomy as described in [7] - in Tali's case, the tube is not present

In this form, the tracheostomy also completely prevents patients from speaking, as the air exhaled from the lungs exits the throat through a tube before reaching the larynx, where speech is produced. This issue, however, has a standard solution in the form of speaking valves, which are attached to the tube and close it during exhalation, forcing the air to pass through the larynx [7]. Users of these valves are capable of producing clearly understandable speech[2].

[1]Which is limited as we did not get access to her medical documentation.
[2]https://www.youtube.com/shorts/EJ6paPekh40

Clearly, Tali's condition differs in many respects: she has lived with the stoma for many years, does not use a tube, and is capable of producing speech. Therefore, in her case, the terms *long-term tube-free tracheostomy* [9] or *persistent tracheal stoma* [10] might be more medically accurate. However, since this is not a medical project and Tali's most prominent speech impediment stems from having a stoma in the trachea, we consider it appropriate to use the simplified term *tracheostomy*.

Furthermore, impaired motor control or general muscle weakness resulting from the neurological damage might be hindering Tali's speech as well.

2.1. Intelligibility Estimation

To estimate the intelligibility of Tali's speech for listeners unfamiliar with it, we asked eight respondents to transcribe twenty-five segments of her speech. Twenty of these segments were sampled from a real-world conversation, and five from a scripted, artificial one (for details on artificial conversations, see Section 4.1.2). Each segment contained between 1 and 7 words, with an average of 2.67.

The audio was played at a high volume, so the errors reported below cannot be attributed solely to the speech being too quiet. Each sample was played exactly once, and there was no time limit for writing the transcriptions. One respondent had prior experience with Tali, having assisted during the recording day (see Section 4.1.3); their results are reported separately.

Participants were given feedback only at the very end of the experiment; therefore, no learning effect could have influenced their performance. The results are presented in Table 2.1.

	At least one	**All**
All respondents	0.128	0.031
Respondents excluding the assistant	0.103	0.023
The assistant	0.28	0.08

Table 2.1.: The relative frequencies of at least one/all words being correctly recognized by a participant for a sample

Although our experiment involved only a small number of participants and listening samples, it demonstrates that Tali's speech is largely unintelligible to untrained listeners and challenging even for an experienced listener (the assistant). In general, the words recognized by the respondents were relatively short. Notably, the only sample in which respondents — other than the assistant — were able to recognize all the words contained just a single word: *Ne* ("No" in English).

Moreover, participants showed considerable confusion and helplessness during the experiment, and fatigue in its later stages.

These results suggest that an ASR system could improve Tali's life drastically. However, the low performance of even a trained assistant indicates that recognizing Tali's speech will likely be challenging for deep learning models as well.

3. Related work

To our knowledge, the scientific literature offers no precedent for using state-of-the-art deep learning ASR methods as an assistive tool for individuals with permanent tracheostomy. Consequently, the related work for this thesis draws from two distinct fields with limited exchange of ideas: deep learning-based ASR and assistive technologies for people with speech disabilities. To account for their limited overlap, we dedicate a separate section to each of these topics.

3.1. Deep learning ASR

As the ways of uttering a sentence largely vary among and even within individual speakers, ASR has been, from its very beginnings, heavily reliant on statistical and ML methods. This is evidenced by the reasoning of Davids et al. - inventors of the first ASR system (speaker-dependent spoken digit recognition) in the 1950s: *Since design of any successful recognition circuit demands a quantitative knowledge of an inherently variable speech signal, any useful description of this signal must be expressed in statistical terms.* [2]

The authors of the aforementioned work computed frequency statistics of individual digits (as pronounced by a single speaker) and then hardwired those into their ASR circuit. This template-based approach has been prevalent in the field also for the following decades [11]. The shift to ML approaches came in the 1970s and 1980s [12]. At that point, hidden Markov models (HMMs) became the standard algorithm for ASR [13] and have been widely used in practice till today.

The 21st century brought increased focus on deep neural networks (DNNs). In 2008, Graves et al. proposed Connectionist Temporal Classification (CTC) [14], an RNN-based ASR system that slightly outperformed both HMM and hybrid HMM-DNN systems on a dataset of phonetically segmented and annotated English speech. The critical challenge of this approach was mapping the RNN's frame-level predictions to a sequence of discrete labels, which was typically much shorter. To address this misalignment between predictions and targets, the authors introduced the *CTC loss* function, which uses a forward-backward algorithm to compute the likelihood of a valid label sequence given the model's outputs. Their approach relies on a simplifying assumption: that the model's outputs are conditionally independent, given the internal state of the network. For further details, we refer interested readers to the original paper.

In the following years, the researchers aimed to develop *end-to-end* neural ASR methods that would mitigate shortcomings of CTC (i.e. its conditional independence assumption and non-trivial training and inference algorithms). A particularly important innovations were *sequence-to-sequence learning* [15] and *attention* [16].

Sequence-to-sequence learning eliminates the need to explicitly align the input sequence with target labels. Instead, a dedicated RNN, the *encoder*, processes the input and produces an embedding, which is then passed to another RNN, the *decoder*, that generates the final transcription (as depicted in Figure 3.1). This architecture enables *end-to-end* ASR, as demonstrated by Sak et al. in the Recurrent Neural Aligner [17].

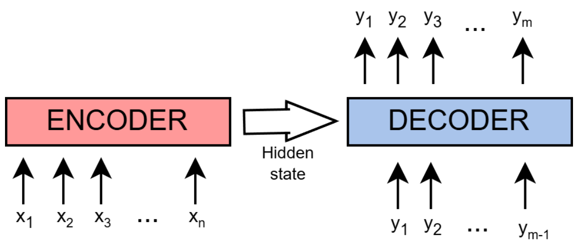

Figure 3.1.: Sequence-to-sequence architecture

Sequence-to-sequence learning as proposed in the original paper, however, has the caveat of using a fixed-size hidden state, which limits its capability of processing longer sequences. To overcome this problem, Bhandau et al. [16] utilise the *attention* mechanism. Their architecture comprises an *alignment model* that learns how is each of the input tokens is relevant for each of the output ones. We omit the computational details as the *attention* mechanism has many variants and their rigorous survey would suffice for a standalone thesis. As an example of early use of *attention* for ASR, we refer an interested reader to *Listen, Attend and Spell* by Chan et al. [18].

Sequence-to-sequence learning and *attention* are jointly used for ASR by Google in 2017 [19] and for the first time outperform the classical HMM approaches on Google's dataset of

12,500 hours of spoken English with noisy background. Perhaps even more notably, they serve as core mechanisms of a general-purpose *transformer* [20] architecture that omits the recurrent component of the classical *sequence-to-sequence* architecture and thus allows highly efficient parallelisation.

The current state-of-the-art models generally use the algorithms presented above and achieve low error rates thanks to a large number of parameters. Below, we present the current open-source state-of-the-art models (i.e., those we considered for our own project) in detail.

3.1.1. DeepSpeech

Although the DeepSpeech [5] model dates back to 2014, it still achieves competitive performance for the Czech speech[1] (15% WER for Common Voice dataset [21]) among models that are sufficiently light-weight for real-time inference.

In the original paper, Deep Speech takes spectrograms as input and outputs character-level transcriptions (including a blank symbol required by the CTC loss). It is a bidirectional RNN that, at each timestep, processes a temporal bin of the spectrogram. The original Deep Speech model consists of five layers: the first three operate on individual temporal bins, the fourth is a bidirectional recurrent layer, and the final layer again operates bin-wise.

Unlike the open-source implementation by Mozilla, depicted in Figure 3.2, the original paper used a vanilla RNN rather than an LSTM for the recurrent layer. The authors later introduced Deep Speech 2 [22], which includes up to seven recurrent layers.

[1] https://github.com/comodoro/deepspeech-cs/releases

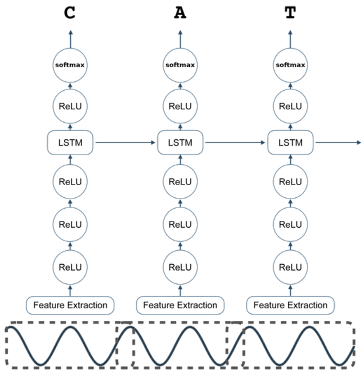

Figure 3.2.: DeepSpeech architecture as implemented by Mozilla[3]

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Deep Speech is trained using the *CTC loss function*. It is commonly paired with an n-gram language model, which helps guide the transcription toward existing vocabulary in the language. Specifically, the final transcription maximises the following objective function:

$$Q(\text{trans.}) = \log P_{ASR}(\text{trans.}|\text{sound}) + \alpha \log P_{LM}(\text{trans.}) + \beta \,\text{word\_count}(\text{trans.}) \quad (3.1)$$

where $P_{ASR}(\text{trans.}|\text{sound})$ stands for transcription probability as estimated by the DeepSpeech and $P_{LM}(\text{trans.})$ for transcription probability as estimated by the n-gram. α and β are standard hyperparameters. The authors of the original paper maximised the objective with a beam-search of beam size 1000 - 8000. For the aforementioned Czech subset of the Common Voice data, the DeepSpeech achieved almost three times lower WER when integrated with n-gram (15% with vs 41% without)[2].

DeepSpeech has been implemented and provided as open-source under a public license by Mozilla[3]. The only difference from the original solution was the use of LSTM instead of RNN in the recurrent block. DeepSpeech is also provided by *Torchaudio*[4].

[2]Results are reported in an unofficial GitHub repository maintained by an independent AI enthusiast: https://github.com/comodoro/deepspeech-cs/releases. While not a formally peer-reviewed source, it is currently the only available reference for these specific results.
[3]https://deepspeech.readthedocs.io/en/r0.9/DeepSpeech.html
[4]https://pytorch.org/audio/2.1.1/generated/torchaudio.models.DeepSpeech.html

3.1.2. Wave2Vec

Wave2Vec models [23, 6] have been shown to not only outperform DeepSpeech 2.0 on the WSJ benchmark [24] (various speakers reading aloud articles from the Wall Street Journal) but also perform very well on the Czech speech with Wave2Vec 2.0 achieving 4% WER with 90M parameters [25].

Both the original Wav2Vec [6] and the newer Wav2Vec 2.0 [23] utilise unsupervised pre-training. In the pre-training phase, the contrastive loss is used, and models learn to distinguish between audio segments fitting the provided context and distractor segments. Specifically, the unsupervised objective is to minimise an error function of the following high-level form:

$$\mathcal{L}(\text{cont., sample\_true}) = -\log \frac{\text{sim}(\text{g(cont.), sample\_true})}{\sum_{\text{sample\_distract}} \text{sim}(\text{g(cont.), sample\_distract})} \tag{3.2}$$

where *sim* denotes the similarity function, *cont.* refers to the context provided to Wav2Vec (i.e., the audio surrounding the masked-out target) as input, and g represents the model itself.

After pre-training, Wav2Vec models are fine-tuned using the CTC loss in a supervised manner. It has been shown that, when pre-trained on vast amounts of unlabelled data (on the order of thousands of hours), they require only a few hours of labelled data to achieve state-of-the-art performance in a given domain—and only a few minutes to reach decent, albeit suboptimal, results [23].

Similar to Deep Speech, Wav2Vec models use a sub-sentence language model during inference to guide predictions toward existing vocabulary [23, 6, 25]. When used for transcription, individual Wav2Vec outputs correspond to individual characters.

Apart from the aforementioned principle, Wav2Vec [6] and Wav2Vec 2.0 [23] are largely different. Wav2Vec is a fully convolutional network [26] that uses Mel spectrograms as inputs whereas Wav2Vec 2.0 combines convolutional layers with transformer ones and takes raw waveform as input (as depicted in Figure 3.3). There are also notable differences in the implementation of the unsupervised pre-training, but for the precise details, we refer the interested reader to the respective papers.

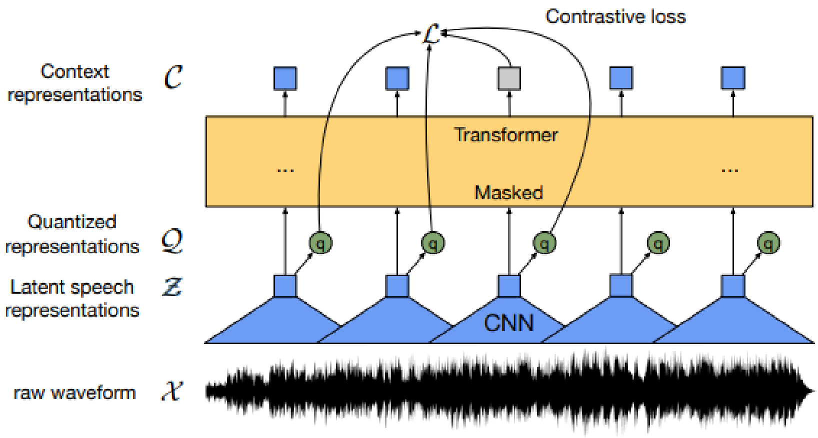

Figure 3.3.: Wave2Vec 2.0 architecture - taken from the respective paper [23]

Pre-trained Wav2Vec is publically available via HuggingFace both for English[5] and for Czech [12].

3.1.3. Whisper

Whisper [1] is a relatively recent (2022) ASR model that exhibits strong zero-shot performance even on out-of-domain data. It is provided in seven versions of five different sizes of which the larger ones generalise to healthy Czech speech already quite well (large-v3 achieves zero-shot 9% WER on Czech Common Voice[6]). However, also for the smaller models, many fine-tuned versions exist on Hugging Face and claim high performance[7,8,9].

[5] https://huggingface.co/facebook/wav2vec2-base
[6] https://github.com/openai/whisper
[7] https://huggingface.co/mikr/whisper-small-cs-cv11
[8] https://huggingface.co/LadislavVasina1/whisper-base-cs
[9] https://huggingface.co/kozak-vaclav/whisper-tiny-cs

Whisper is a standard encoder-decoder Transformer [20] that takes Mel spectrograms as input. Byte Pair Encoding (BPE) [27] is used to tokenise the text. The spectrogram patches are first processed by a small CNN with a GELU activation function [28], after which sinusoidal positional embeddings are added. The decoder, in contrast, uses learned positional embeddings. We depict the architecture in Figure 3.4.

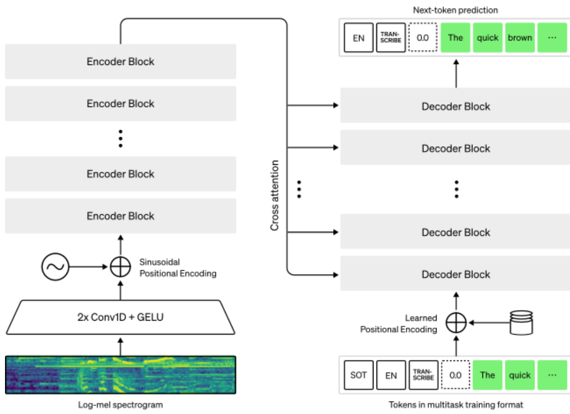

Figure 3.4.: Whisper architecture - taken from the OpenAI website[6]

The main innovation behind the Whisper model lies not in its architecture or training method, but in the vast amount of data used for training. Unlike the previously mentioned models, which were trained on relatively limited datasets with human-validated labels such as Common Voice [21], Whisper was trained on 860,000 hours of speech, reportedly human-annotated but not manually validated, scraped from the internet. Outputs from other ASR systems were filtered out from this dataset.

Whisper is provided in a dedicated Python library[6]. Furthermore, many fine-tuned versions are available via Hugging Face[7,8,9].

3.2. Alternative and Augmentative Communication

The term Alternative and Augmentative Communication (AAC) refers to techniques that help people with various communication impairments and disabilities to cope with their handicaps [29].

In the literature, AAC is categorised into the following two classes: low-tech and high-tech [30].

As *low-tech AAC* we typically understand approaches that do not require an external power source in order to work. Low-tech methods include writing the messages down, pointing to letters on a board, communication cards, sign languages, eye-blink codes, etc. [31]

High-tech AAC refers to more sophisticated approaches that typically utilize an electrical device and require a power source to operate. Traditionally, they include tools that read aloud, what the user has written [32] or recognise a sign language [33]. ASR systems represent another typical high-tech approach [31].

The AAC ASR systems typically utilise the very same machine-learning algorithms as ASR systems for the healthy speech - HMMs [34], LSTMs [35], state-of-the-art ASR architectures as Wave2Vec 2.0 [36] or deep autoencoder for feature preprocessing [37]. Unfortunately, the relevance of these works was limited for our own project, because they largely focused on patients affected by dysarthria (neurological damage commonly caused by stroke or Parkinson's disease) and not the peculiar condition of Tali.

As the WER of AAC ASR systems is typically very high, a common approach is to limit the vocabulary to only a set of words. As an example, we refer to the work of Hawley et al. [34] who developed an AAC ASR system based on HMMs (VIVOCA) for seven people with severe dysarthria. The system recognises only 10 - 50 words for every patient, and these words are then used to navigate via a tree of predefined sets of phrases like 'I would like a cup of coffee' or 'Could you open the window please?' (as depicted in Figure 3.5). When the phrase is selected, an artificial voice reads it aloud. The system achieves over 90% accuracy for all but one patient. However, it received criticism from users for being too slow and for the vocabulary being limited too much.

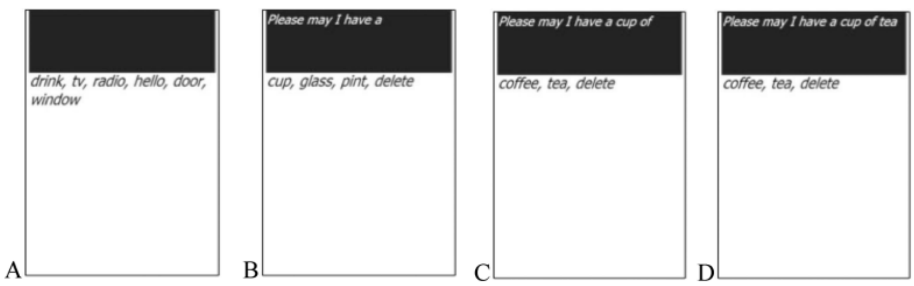

Figure 3.5.: VIVOCA user interface [34]: on the highest level A the user chooses to ask for a drink, they are directed to level B where they specify that they want a cup of the drink, on level C they specify that the drink in the cup should be tea. When the whole phrase is selected (D), the system reads it aloud.

Recently, brain2speech AAC systems were introduced. In particular, Card et al. [38] developed AAC for a patient with severe dysarthria caused by ALS that used neuroprosthetics to collect the data from the patient's brain. The system achieved 2.5% WER with only 16.6 hours of training data. For this purpose, the authors used four machine learning models to predict phonemes from the brain waves, to predict sentences from the phonemes, to choose the most appropriate sentence and to generate an artificial patient's voice saying the sentence out loud. We deem our own work to be relevant even in the context of these advancements, as our method is non-invasive and less expensive.

4. Data Collection

For training, Whisper [1] requires audio segments of speech (up to 30 seconds long) paired with their corresponding transcriptions. Moreover, the transcriptions do not need to be temporally aligned with the audio at the word or subword level. For our specific case, we furthermore focus on collecting conversational data as the target domain ouf our system are Tali's conversations with other people.

Given that understanding Tali's speech is challenging even for trained listeners, we aimed to collect as much data as possible. Specifically, our goal was to obtain at least twenty hours of annotated speech from her. We consider this amount sufficient, based on the results of Eckert and Schuppler [39], whose Whisper-based ASR system for a child with dysarthria achieved useful performance after being trained on less than ten hours of data.

We conducted over one hundred recording sessions and annotated more than twenty hours of Tali's speech. This was made possible primarily through *artificial conversations*—a novel and efficient approach to data collection—which we describe in detail below, along with other methods we employed.

4.1. Data Collection Methods

We applied the following three methods for data collection. We examine each of them in detail below.

- **Reading sessions**
- **Artificial conversations**
- **Recording day**

4.1.1. Reading sessions

Reading sessions were the initial method used for data collection. Tali's task was to read a page of text of her choice (typically an article from a popular space magazine) and then send us both the scan of the page and a video recording of her reading.

During post-processing, we extracted text from the scan and split it into individual words by whitespaces. We then split the recording (MOV) into video (mp4) and audio (mp3). Both the original and the resulting audio had a sample rate of 44.1 kHz with 32 bits per sample. For audio we detected all the samples with amplitude over 800 PCM (determined experimentally) and stored the segments from 0.25s before the peak to 2s after the peak. Finally, we manually aligned the segments of audio to the pieces of text (mostly excerpts of 1 - 3 words).

As Tali's speech is highly unintelligible, and she has a tendency to modify the provided scripts, we found our annotation process to be highly error-prone. To assess its quality, we developed a web interface[1] (depicted in Figure 4.1) for verifying and, if necessary, correcting the labels. We then asked Tali's mother to use this tool to review two reading sessions, comprising approximately 250 samples in total. Based on her evaluation, we estimate that only about 2–3% of the samples from the reading sessions are mislabelled.

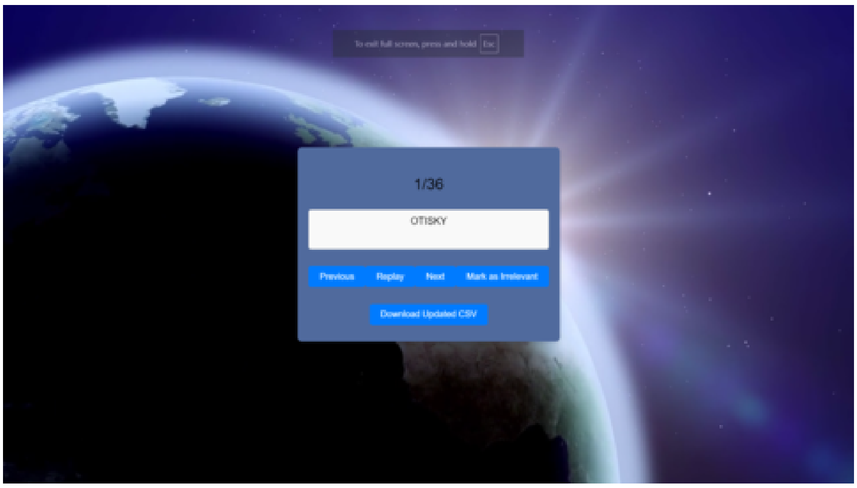

Figure 4.1.: The interface for annotation verification

[1] http://david.pruzor.cz/David/check_annotations/casopisy.html

Although we found the error rate of 2-3% reasonably low, the reading sessions were nevertheless soon abandoned for the following reasons:

1. As post-processing a 15-minute session, typically requires around 90 minutes of highly concentrated work, it was unfeasible to collect the 20 hours of data, which we have set out as our goal, this way.
2. The vocabulary used in the space magazines is not representative of our target domain - real-world conversations.
3. The pace, tone and other non-verbal aspects of speech probably also differ significantly between reading and real-world conversations.
4. It was difficult for Tali and her family to keep their motivation high, as the reading sessions turned out to be monotonous.

4.1.2. Artificial conversations

The predominant type of recording sessions was *artificial conversations*. For these, we provided Tali with an audio file containing a dialogue between characters A and B. Her task was to repeat the lines spoken by character A. The structure of artificial conversations is illustrated in Figure 4.2. She recorded herself during the procedure, and upon completion, she sent us the video recording.

Figure 4.2.: Artificial conversations - the upper line depicts the structure of the audio file prepared by us. The lower line displays at what moments we expected Tali to speak.

The process of generating the artificial conversations was loosely inspired by RefGPT [40], a method proposed for generating truthful dialogues using LLMs. We prompted ChatGPT

(gpt-4o-mini) [41] with the beginning of the dialogue, 1–4 sentences describing its high-level topics, detailed instructions on the manner in which the characters should speak, and occasionally also a source article on the conversation's topic (the prompt is provided in Appendix in Section A). As an example, we provide an excerpt of a conversational script in Figure 4.4.

The resulting dialogue was split into lines spoken by character A and lines spoken by character B. Both sets of lines were converted to audio using Google Text-to-Speech[2]. The lines spoken by character B were further processed with voice conversion to make them distinguishable from those of character A. For this purpose, we typically used the Voice.ai service[3], though we occasionally opted for Seed-VC [42] to enrich the conversations with the voice of a publicly known person, such as—but not limited to—Elon Musk.

Finally, we merged the audio into a single dialogue file, which also included an intro and outro song excerpt (generated by Suno.ai[4]) as well as a recap of the instructions. While creating the audio file, we also saved the temporal coordinates of the silence segments during which Tali was expected to speak, along with the corresponding lines of text. These timestamps were later essential for processing the recordings efficiently.

We visualise the procedure described above in Figure 4.3.

As already mentioned, we provided Tali with 10 seconds of silence to repeat each of character A's lines. In addition, we later included a 20-second silent segment after every 40 dialogue lines (spoken by both characters). During these longer pauses, Tali had the opportunity to freely comment on the dialogue from her own perspective. Although we did not annotate these segments, they served an important role: they allowed Tali to express her conversational agency, which might otherwise have interfered with her ability to precisely repeat the prescribed lines. We also preserved these segments for potential future use in semi-supervised or unsupervised training.

[2] https://cloud.google.com/text-to-speech
[3] https://voice.ai/tools/voice-changer
[4] https://suno.com/

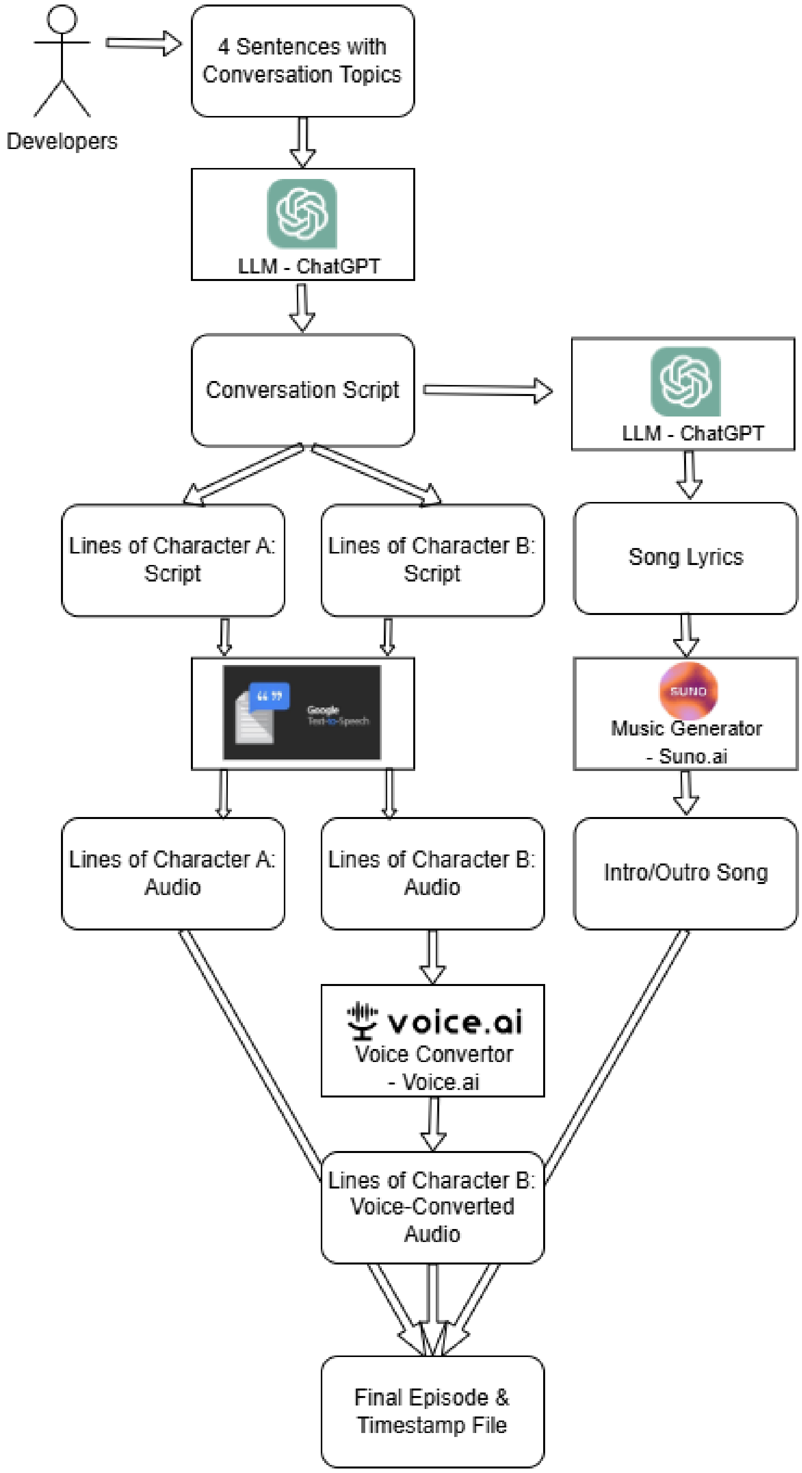

Figure 4.3.: Artificial conversations - scheme of the generation process

In order to make the artificial conversations both engaging and representative of Tali's real-world vocabulary, we dedicated most of the episodes to topics such as Tali's friends, home city and hobbies. We also made episodes on various everyday topics such as school or family relationships. We also kept the lines short to mirror her real-world, concise manner of speech. We provide an example of a conversation script in Figure 4.4. Furthermore, a full script of an artificial conversation can be found in the Appendix in Section B.

Tomka: I have another idea.
Shalina: What idea?
Tomka: I want to give a hundred parrots as a gift.
Shalina: That's a big number! Why parrots?
Tomka: Martin ***** likes birds.
Shalina: That's right, he loves birds.
Tomka: I think they'll make his family happy.
Shalina: Where are you going to get them?
Tomka: We have them at the zoo.
Shalina: Who will give them to you?
Tomka: I want all 100 of them.
Shalina: What will you do with the old ones?
Tomka: I'll choose them carefully.
Shalina: What kind of parrots do you want?
Tomka: Mostly colourful and intelligent.
Shalina: That sounds great!
Tomka: Maybe even a few rare ones.
Shalina: Do you know when to donate yet?
Tomka: For Martin's Day, maybe.

Figure 4.4.: Conversation Excerpt. Tomka is the character A. Shalina is the character B. We masked out the family name of Tali's real-world friend, Martin. The excerpt is translated from the original Czech to English.

We post-processed the recordings analogously to those from the reading session. We again split MOV files into mp3 and mp4 ones and then aligned the audio with the text. The task was however considerably easier than with the reading sessions, as we had already stored the transcriptions and temporal coordinates of the corresponding audio segments. The only necessary task was to align the start of Tali's speech with the moment Tali started to talk, and the rest could have been computed from that using the information saved during conversation creation. However, we still manually inspected the segments to verify the collected data.

In conclusion, we collected a large majority of the data (for exact numbers see Chapter 5) using this method as it was engaging for Tali, fast to process and as representative of the real-world conversations as we were able to get.

4.1.3. Recording day

When discussing Tali's case with professional phoneticians, we were advised to make some recordings in a professional studio to ensure a low signal-to-noise ratio (SNR). To this end, we organised a recording session at the *Recording Studio of Nad Orlí Theater in Brno*.

During the recording day, we captured audio only. Whenever Tali began to speak, we recorded the timestamp using a simple dedicated program. In the case of real-world conversations, we also manually transcribed her speech in real time. This was made possible primarily by the presence of Tali's mother, who translated most of her utterances, as well as by two assistants familiar with Tali, who supported the transcription process, time-stamping, and helped keep the conversations engaging.

During the recording day, we captured audio only. Whenever Tali began to speak, we recorded a timestamp using a simple, dedicated program. For the real-world conversations, we also manually transcribed her speech in real time. This was possible thanks to the presence of Tali's mother, who translated most of her utterances, and to two assistants with prior experience working with Tali, who supported the transcription, time-stamping, and helped keep the conversations engaging.

In postprocessing, we aligned the first moment of Tali's speech with the first timestamp. We then manually aligned audio segments around those timestamps with the transcriptions taken during the recording day.

5. Data Analysis

Our design choices were largely motivated by the specific characteristics of Tali's voice. Therefore, we find it appropriate to mention these factors as an explanation and justification for some of the techniques described in Chapter 6.

5.1. Qualitative Data Analysis

After listening to several minutes of Tali's speech and looking and corresponding spectrograms, several irregularities come out clearly. We illustrate them in the Figure 5.1.

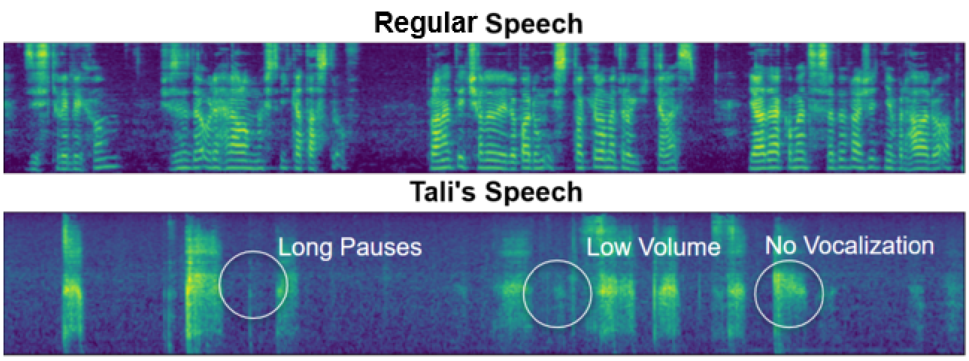

Figure 5.1.: Mel spectrograms (128 bins) of regular and Tali's speech

1. **Absence of vocalization**: The harmonic component of the speech is missing. In the plots above, this is demonstrated by the absence of parallel horizontal stripes that are prominent in the plot of regular speech. However, unlike in the provided plot, Tali occasionally vocalised the first vowels of words, especially when they occurred at the beginning of sentences and the respective sessions.

2. **Long pauses**: Tali's speech is slow by itself, but her rate of speech is further slowed down by long pauses between and within individual words. These pauses can be as long as one second (as shown in the plot above). Especially the latter type represents a great challenge for automatic word detection and, consequently, for the ASR system.

3. **Low volume**: The somewhat lower contrast of Tali's spectrogram compared to that of healthy speech indicates the weakness of her voice. This poses a challenge for our system because everyday noises, such as a car driving on the street, usually drown out Tali's voice. (As an alternative example, the verification of data annotation described in Section 4.1.1 led Tali's mother to move their turtle terrarium, as she found that the turtle's trampling significantly interfered with Tali's speech.)

Other notable Tali's speech defects encompass a tendency to sometimes repeat vowels or a slight voice tremble. However, we have identified the above as the most serious.

5.1.1. Expert Analysis

We have also reached out to expert phoneticians (As. Prof. Radek Skarnitzl and Dip. Ing. Tomáš Bořil, PhD from Institue of Phonetics, Philosophical Faculty, Charles University in Prague) and asked them to analyse Tali's speech as well. Their findings mostly correspond to ours:

1. **Periodic source of speech is missing**: This observation builds upon the source-filter model of human speech [43]. In the source-filter approximation of speech production, speech is viewed as a combination of periodic and noise signals produced by the vocal folds, which are then modulated by linear filters implemented by the oral and nasal cavities. Tali's most critical problem was found to be the absence of the periodic component. We illustrate this issue in Figure 5.2. This expert finding corresponds to the same phenomenon we previously referred to as *Absence of vocalisation*.

2. **Low SNR**: The phoneticians noted that Tali's speech is rather quiet and easily overwhelmed by surrounding noise. For this reason, they also recommended conducting some recordings in a dedicated recording studio.

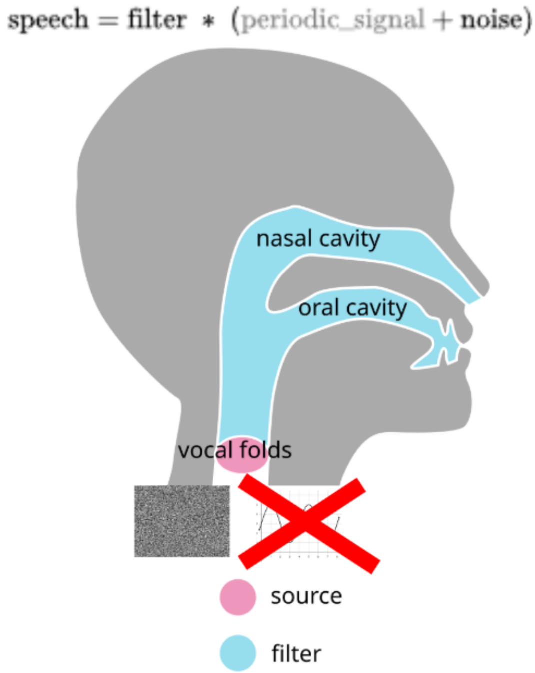

Figure 5.2.: Source-filter model of speech production: we highlight the missing periodic component of Tali's speech (the image taken from Wikipedia[1]).

5.2. Quantitative Data Analysis

In Table 5.1, we also provide some quantitative attributes of our dataset. We provide these values separately for data of every collection method (i.e. *Reading sessions*, *Artificial conversations*, *Recording day*). Furthermore, we provide them for *Real-world conversations* (even though they are a subset of *Recording day*) because they play a prominent role in the evaluation of our system.

For each of these subsets, we calculate the number of recording files, total recording time (i.e., the total length of video files submitted by Tali), total annotated time (i.e. the

[1] https://en.wikipedia.org/wiki/Source-filter_model

actual length of the audio used for our experiments), number of samples and number of words used in annotations. From those, we also compute average words per minute (for comparison: Common Voice Czech speakers use 99 words per minute on average) and words per sample.

Furthermore, we provide relative spectral flatness [44] as a measure of how noisy the recordings are[2]. We measure this quantity in multiples of the spectral flatness of a healthy voice recorded using a laptop microphone.

	Reading Sessions	Artificial Conversations	Recording Day	Real-world Conversations
Number of Recordings	21	69	12	2
Total Recorded Time (h)	4:08	38:13	3:00	1:31
Annotated Time (h)	3:06	20:21	1:48	0:32
Number of Samples	2 719	8 428	781	327
Number of Words	4 739	31 499	3 273	1 159
Words per Minute	25.4	25.8	30.4	32.2
Words per Sample	1.7	3.7	4.2	3.5
Relative Spectral Flatness	1.33	1.3	8.1	1.6

Table 5.1.: Overview of the recorded audio dataset. The times are provided in *hours:minutes* format.

We provide most of this data as a public dataset via Zenodo[3]. We excluded the segments clearly mentioning Tali's associates.

[2]Spectral flatness is computed for every time step of a spectrogram as $\frac{\text{geometric mean of frequency bins}}{\text{arithmetic mean of frequency bins}}$. We aggregate over all frames of each sample and then again for all the samples using the arithmetic mean.
[3]https://zenodo.org/records/15225595

6. Training pipeline

We use Whisper by OpenAI [1] as it is a state-of-the-art architecture that is also computationally feasible to train (at least in the smaller versions), well supported by a dedicated Python library[1] and plenty of pre-trained and fine-tuned versions exist on the internet[2,3,4], that can be used as a sanity check of our own setup.

On a high level, our training pipeline consists of the following steps:

1. Pre-train Whisper on regular Czech speech.
2. Transform regular Czech speech into quasi-tracheostomy speech.
3. Fine-tune result of step 1 on this quasi-tracheostomy speech.
4. Finally, fine-tune the result of step 3 on Tali's speech.

Below, we provide details on the non-trivial techniques used for training. The exact hyperparameter settings for Whisper experiments and their results can be found in Chapter 7.

6.1. Pre-Training

As collecting and labelling recordings from Tali required considerable effort from her family and us, we decided to pre-train our models on regular Czech speech and also simulated tracheostomy speech.

[1] https://github.com/openai/whisper
[2] https://huggingface.co/mikr/whisper-small-cs-cv11
[3] https://huggingface.co/LadislavVasina1/whisper-base-cs
[4] https://huggingface.co/kozak-vaclav/whisper-tiny-cs

6.1.1. Pre-Training on Regular Czech Speech

Default Whisper models as provided by OpenAI[5] are multilingual and they have also been trained on Czech speech; however, especially when of smaller sizes, they perform suboptimally and benefit from being fine-tuned.

Although, there are multiple fine-tuned Whisper models on Hugging Face[6,7,8] we fine-tuned them ourselves which was a useful sanity check of our Whisper training loop.

We used Common Voice CS 19.0 [21, 45] for the fine-tuning. This dataset provided us with 5.5 GB of recordings of 266 hours of speech. 78 of those 266 hours were verified. Both collection and verification were crowd-sourced by 1 023 speakers. In a questionnaire for the speakers, 54% of them filled in to be male while only 22% were female. Furthermore, 47% of speakers filled in to be 30 - 39 years old, and another 18% filled in age 20 - 20. Only 2% contributors filled in an age under twenty, which is the group Tali belongs to.

6.1.2. Pre-Training on Quasi-Tracheostomy Speech

Furthermore, we attempt to simulate Tali's impediments and turn regular Czech speech into a quasi-tracheostomy one. We formalise this problem using the following formula:

$$t^* = \arg\min_t D\left(p_{t(X_{\text{regular}})} \parallel p_{X_{\text{Tali}}}\right) \tag{6.1}$$

Here, t denotes a transformation function and D represents a divergence measure between the distributions of transformed regular speech and Tali's speech. In our implementation, rather than explicitly defining and minimising a divergence, we instead maximise a similarity measure.

[5] https://github.com/openai/whisper
[6] https://huggingface.co/mikr/whisper-small-cs-cv11
[7] https://huggingface.co/LadislavVasina1/whisper-base-cs
[8] https://huggingface.co/kozak-vaclav/whisper-tiny-cs

Our custom similarity function receives two sets of audio files and then computes the score as follows:

1. Computes 512-dimensional voice embeddings of all the audios using voice embedding by pyannoteAI [46].
2. Clusters the data to two clusters with K-Means.
3. For each cluster, it computes its inner entropy of original classes (i.e. quasi-tracheostomy and tracheostomy).
4. Similarity is then computed as the mean of these two entropies.

Note that this similarity achieves its minimum when even after transformation, the voice embeddings are so distinct that K-means separates them perfectly which would lead to entropy 0[9]. On the other hand, if K-means completely fails to copy the original classes, entropy will be 1[10].

By manual experimenting and local hill-climbing that refined its results, we found the following combination of transformations as the most helpful:

1. **Removal of the periodic voice component**: Inspired by the phoneticians' insights regarding the missing periodic source in the source-filter model, we removed it from the regular speech as well. To achieve this, we used a script they provided, which applies linear predictive coding [47] to isolate the periodic component and replace it with white noise (we provide the script in the Appendix in Section C).
2. **Time stretching**: We slowed down the regular speech by a factor of four. This factor was chosen to approximate Tali's speaking rate, which is roughly four times slower than that of typical speakers (25–30 words per minute for Tali vs. 100–150 words per minute for regular speech).
3. **Pause insertion**: To simulate Tali's tendency to insert pauses within her speech, we added seven silent segments of 250 ms each at randomly sampled time points throughout the recordings.

[9] $\mathbb{E}[-\log_2 X_i] = -\sum p(X_i) \log_2 p(X_i) = -1 \log_2 1 = \mathbf{0}$
[10] $\mathbb{E}[-\log_2 X_i] = -\sum p(X_i) \log_2 p(X_i) = -(0.5(-1) + 0.5(-1)) = \mathbf{1}$

We tested the effectiveness of these preprocessing steps on 60 one-sentence-long recordings of regular speech and another 60 recordings of Tali's speech. Before any transformations were applied, the intra-cluster entropy was 0.335. After preprocessing, it increased to an average of 0.781. Spectrograms of regular, quasi-tracheostomy, and Tali's speech are shown in Figure 6.1.

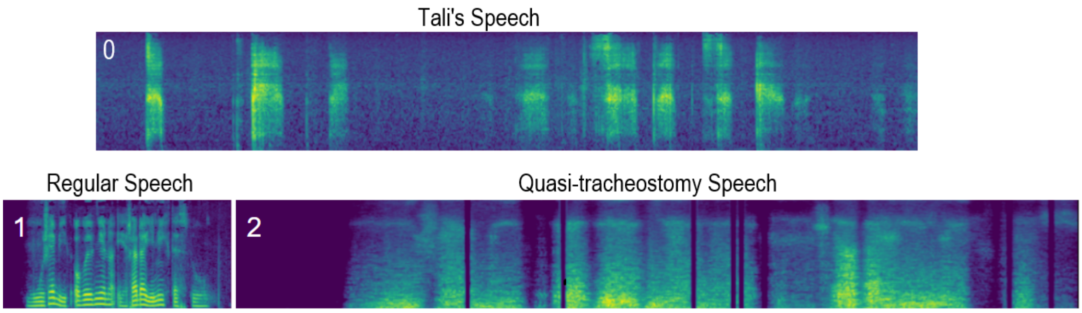

Figure 6.1.: Mel spectrogram (128 bins) of a regular voice before (1) and after (2) the transformations. Also, we provide a spectrogram of Tali's speech (0).

6.2. Knowledge Distillation from MLM

After inspecting transcriptions produced by Whisper Tiny, we found that even when provided with the ground truth of all preceding tokens, it often predicts words that not only do not exist in the Czech vocabulary but are also hardly pronounceable. To mitigate this issue, we attempted to improve Whisper's linguistic understanding through knowledge distillation from a masked language model (MLM).[11] We provide a visual intuition in Figure 6.2.

For our language model, we used a neural network consisting of an embedding layer, six bidirectional LSTM layers [48], and layer normalization [49], totalling 60 million parameters. We applied the same byte pair encoding (BPE) [27] tokenisation as Whisper, which was necessary to compare the outputs of our language model with Whisper's predictions. As training data, we used the CC100-Czech dataset [50], which contains 4.4 GB of Czech text crawled from the internet. However, due to limited computational resources, we used only a 210 MB subset, comprising approximately 1.3 million sentences.

[11] Masked language models (MLMs) are models that take a sentence with one or more tokens masked out and predict a probability distribution over the vocabulary for each masked token.

We trained the bi-LSTM model to predict masked tokens in these sentences and achieved a 93% top-5 validation accuracy after 29 epochs.

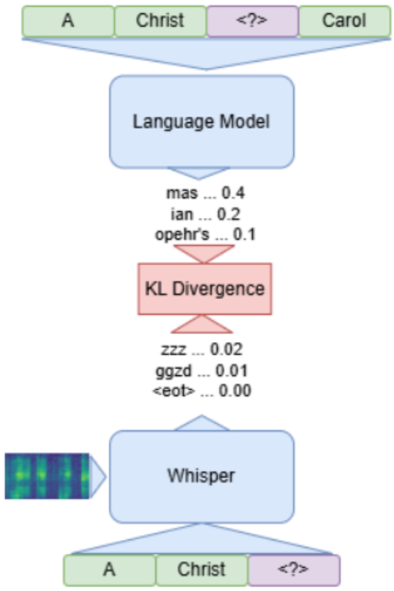

Figure 6.2.: Schematics of KD from MLM to Whisper

We then included this model to our Whisper pipeline as a KD teacher. For every transcription token $\mathbf{t}_t$ the loss was defined as:

$$L_t(\text{asr}, \text{mlm}, \mathbf{t}_{1:T}, \mathbf{s}) = \lambda_{lm} CE(\text{asr}(\mathbf{t}_{1:t-1}, \mathbf{s}), \mathbf{t}_t) + (1 - \lambda_{lm}) KLD(\text{asr}(\mathbf{t}_{1:t-1}, \mathbf{s}), \text{mlm}(\mathbf{t}_{1:t-1 \cdot t+1:T})) \tag{6.2}$$

Here, *asr* denotes the Whisper model, *mlm* the MLM teacher, CE the cross-entropy loss, KLD the Kullback–Leibler divergence loss, $\mathbf{t}_{i:j}$ a sequence of transcription tokens from position i to j (with $\mathbf{t}_{1:T}$ representing the complete transcription), and $\mathbf{s}$ the spectrogram of the recording.

We used KD from MLM for pre-training on both regular and quasi-tracheostomy Czech speech and for Tali's speech.

In Figure 6.3, we provide an example of transcriptions with and without KD from MLM. Although there are plenty of errors in both, this figure already demonstrates a slight .

Sample 1

R: Na této silnici jsou dvě autobusové zastávky.

O1: V této silnici jsou dvě od mousové veávky.

O2: Na této vněi jsou dvě od nemusléé zastávky

Sample 2

R: Což ale platí lidem originálnosť!

O1: Což ale platí lidem originál..?

O2: Což ale platí lidem originální.?

Sample 3

R: Aféra vyvolala řadu demonstrací po městě a musela zasahovat policie.

O1: Aér vyvolala př adu drí o těstě a musela zaáovat policie.

O2: Agh titvolala př�adu devrací po městě a musela zachovat policie.

Sample 4

R: Dalších dvacet mužů bylo vážně zraněno.

O1: Dalšíím vvaaceti mu žů bylo vášné.raně.

O2: Dalšíím dvaceti mžů bylo vášně v raděno

Sample 5

R: Písmena jsou v takovém případě dvakrát vyšší.

O1: Jiímena jsou de krové vípaděpakvakrát vyšší.

O2: Tímena jsou tak takovém případě ivarát vyšší.

Sample 6

R: Architektem byl Percy Allan.

O1: Architektem byl pen Six Suite

O2: Architektem byl pes Al.

Figure 6.3.: Transcriptions with and without KD from MLM. For every sample, **R** stands for the true transcription, **O1** for transcription without KD and **O2** for transcriptions with KD. As Whisper is not constrained to predict only existing words, there are many non-existent ones in both cases. They are marked in red. Also, the very apparent misfits (i.e., words that exist but do not occur in this context) by green.

Note that these transcriptions were produced during validation runs of the respective Whisper training loops, where each token was predicted with access to its true preceding tokens. This setup is fundamentally different from real-world transcription, where the context is composed of the model's own previous predictions.

This prediction strategy likely further amplifies the number of non-existent words, as Whisper often begins predicting an incorrect word but, when provided with the true preceding tokens, switches to the correct target. However, the previously predicted tokens are still printed, resulting in non-existent words.

7. Experiments

In this chapter, we provide the exact hyperparameter settings and the results of our experiments with Whisper [1]. We focus on the following three stages:

1. Regular Czech speech
2. Quasi-tracheostomy speech
3. Tali's speech

We report results for three Whisper model sizes: Tiny (39M parameters), Base (74M), and Small (244M). Additionally, we evaluate two training pipelines: *Baseline* and *Adapted*. The Baseline pipeline does not incorporate knowledge distillation (KD) from the masked language model (MLM) and pre-trains Whisper only on regular speech. In contrast, the Adapted pipeline includes KD from the MLM and also pre-trains Whisper on quasi-tracheostomy speech. Due to limited computational resources, we did not train the Baseline version of the Whisper Small model.

Throughout the experiments, we tuned the learning rate, batch size, and λ_{lm}, while keeping all other hyperparameters fixed across model sizes and pipelines. We applied a weight decay of 0.01 (excluding biases and layer normalisation weights) and used a temperature of 2 for all knowledge distillation (KD) experiments. All models were trained with the AdamW optimizer [51], using $\epsilon = 10^{-8}$ and a linear learning rate scheduler with warmup during the first 8% of total update steps. Each model was trained for 14 epochs.

7.1. Regular Czech Speech

7.1.1. Hyperparameters

For pre-training on Common Voice [21], we used the hyperparameters in Table 7.1. The hyperparameters are optimised for the respective experimental setups.

Model	Learning Rate	KD Lambda	Batch Size
Tiny Baseline	5e-4	-	8
Tiny Adapted	1e-4	1e-3	8
Base Baseline	3e-5	-	4
Base Adapted	3e-5	1e-3	8
Small Adapted	3e-5	1e-3	8

Table 7.1.: Hyperparameter settings for regular Czech speech experiments

7.1.2. Results

With these hyperparameters, we achieved the results presented in the Table 7.2.

Model	Validation Loss	WER	CER
Tiny Baseline	1.236	0.447	0.031
Tiny Adapted	0.636	0.345	0.023
Base Baseline	0.819	0.303	0.019
Base Adapted	0.516	0.312	**0.018**
Small Adapted	**0.344**	**0.209**	0.024

Table 7.2.: Results for regular Czech speech experiments

Note that all these results are based on single experiments and that reproducing our experiments might lead to somewhat different numbers. However, the results indicate that scaling up the Whisper improves the performance. Furthermore, KD from MLM improves Whisper Tiny significantly.

We provide our improved Whisper Tiny for regular Czech speech also to the general public via Hugging Face[1].

7.2. Quasi-Tracheostomy Speech

The subsequent pre-training on quasi-tracheostomy data was applied only for the Adapted pipeline, therefore, the Baseline is not listed in the tables.

By quasi-tracheostomy, we understand the regular Czech speech from Common Voice to which we applied preprocessing steps described in Subsection 6.1.2.

7.2.1. Hyperparameters

For the quasi-tracheostomy speech, we empirically selected the hyperparameters listed in Table 7.3.

Model	Learning Rate	KD Lambda	Batch Size
Tiny Adapted	1e-4	5e-2	8
Base Adapted	3e-5	5e-2	8
Small Adapted	1e-5	1e-3	4

Table 7.3.: Hyperparameter settings for regular quasi-tracheostomy speech experiments

7.2.2. Results

With the hyperparameters from above, we achieved the results in Table 7.4 for quasi-tracheostomy speech.

[1] https://huggingface.co/Hobit2002/whisper_tiny_cs

Model	Validation Loss	WER	CER
Tiny Adapted	1.596	0.627	0.049
Base Adapted	1.273	0.556	**0.041**
Small Adapted	**1.000**	**0.488**	0.042

Table 7.4.: Results for quasi-tracheostomy experiments

Interestingly, despite achieving a considerably lower WER, Whisper Small yielded a higher CER than Whisper Base. We hypothesise that this effect is due to differences in batch sizes between the two experiments. During training, all transcriptions within a batch are padded to the same length using the *end of sequence* token. In the validation phase, the model is required to predict not only the actual transcription but also these padding tokens, which are trivial for the model to reproduce. Larger batches tend to have more padding on average, thereby inflating the number of easily predictable tokens and artificially lowering the CER. This may explain the observed discrepancy.

7.3. Tali's Speech

Finally, we provide our settings and results for the experiments on Tali's speech.

7.3.1. Data Split

We use a traditional train, validation, and test set split for our experiments. We distribute data to data splits based on the type of session by which they were collected:

- **Reading session data** is used exclusively for training, as it does not sufficiently represent the target domain to be suitable for validation. These samples are typically short, often containing only a few words, which provides Whisper with an unrealistically limited context. To mitigate this issue, we concatenate neighbouring samples (along with the intervening silence) into segments of at least five seconds in duration.
- **Artificial conversations** are used for training, validation, and final testing, as they constitute the predominant portion of our dataset.

- **Real-world conversations** recorded during the studio session are reserved solely for final testing. We consider these the most representative of the target domain, which consists of spontaneous, unscripted interactions with others.
- **Other data from the recording day** (e.g., singing and reading pre-scripted dialogues) is used only for training, as it does not reflect the spontaneous conversational nature of our target domain.

The resulting data distribution across our splits is summarised in Table 7.5 below. We distinguish between two types of test data: *In-Domain*, consisting of artificial conversations, and *Out-of-Domain*, comprising real-world conversations. This distinction emphasises that real-world conversations were recorded in a different setting and context, and may therefore differ from artificial ones both acoustically and linguistically. Such variation is important to consider, as the model was primarily trained on artificial conversations.

Data Split	Sessions	Samples	Time
Train	74	8 374	16:31
Validation	11	1 418	3:14
Test: In-Domain	5	682	1:56
Test: Out-of-Domain	2	327	0:32

Table 7.5.: Data split for experiments with Tali's speech

Note that session-wise and sample-wise, our train:validation:test ratio roughly corresponds to a common 8:1:1 split.

7.3.2. Hyperparameters

We empirically decide to use the hyperparameters in Table 7.6 for our experiments.

Model	Learning Rate	KD Lambda	Batch Size
Tiny Baseline	1e-4	-	8
Tiny Adapted	1e-4	1e-3	8
Base Baseline	3e-5	-	8
Base Adapted	3e-5	1e-3	8
Small Adapted	1e-5	1e-3	4

Table 7.6.: Hyperparameter settings for regular Czech speech experiments

7.3.3. Results

Finally, we provide the results which we obtained on the final test set. Before the evaluation, the models were trained on both the training and validation sets. In Table 7.7 we state the results for *In-Domain* and *Out-of-Domain* data separately.

Model	Validation Loss		WER		CER	
	In	Out-Of	In	Out-Of	In	Out-Of
Tiny Baseline	0.931	3.279	0.452	0.975	0.059	0.111
Tiny Adapted	0.689	2.376	0.402	0.854	0.037	0.080
Base Baseline	0.681	2.417	0.361	0.831	0.032	0.061
Base Adapted	**0.569**	2.027	0.349	0.817	0.029	**0.063**
Small Adapted	0.574	**1.995**	**0.319**	**0.774**	**0.0284**	0.068

Table 7.7.: Performance of models on different metrics in in-domain (In) and out-of-domain (Out-of) settings.

Note that Whisper is both trained and validated using *teacher forcing* [52]. In this setup, each prediction is made with access to the ground truth of all preceding tokens. While this approach is straightforward to implement and interpret, it tends to inflate performance metrics, since in real-world transcription scenarios, such ground truth information is unavailable and each token must be predicted based on the model's previous outputs. Nevertheless, to the best of our knowledge, *teacher forcing* is the standard approach when training Whisper[2,3] due to being computationally fast, easy to implement and easy to interpret probabilistically.

[2] https://colab.research.google.com/drive/1P4ClLkPmfsaKn2tBbRp0nVjGMRKR-EWz
[3] https://huggingface.co/blog/fine-tune-whisper

7.4. Discussion

Even with the aforementioned caveat in mind, several insights can be drawn from these results. Most notably, there is a substantial gap in performance between artificial conversations (*In-Domain*) and real-world conversations (*Out-of-Domain*). The magnitude of this difference is surprisingly large, suggesting that optimal performance on real interactions will require training on data drawn from those same settings. This presents a significant challenge because such data is difficult to acquire, particularly early on. Annotating recorded audio is expensive, and the precise application settings—and thus the exact target domain—are often only determined through insights gathered throughout the entire development process. Real-world recordings presented several unique challenges: the audio segments often captured only short phrases, typically 2-3 words, and in many cases, the final vowels were cut off. These issues arose because Tali's mother, who served as an interpreter for us, did not always wait until the end of Tali's sentences—or even the end of individual words—before beginning her translation. Since we considered it necessary to remove the mother's voice from the recordings, we were forced to trim the audio, which in turn reduced the contextual information available in each sample.

As shown in Table 7.8, for *In-Domain* data, Whisper Tiny and Base achieved WERs comparable to those observed for regular Czech speech, with the adapted pipelines showing considerable, though not drastic, improvements. The best overall performance was achieved by Whisper Small; however, its results were still significantly worse than for regular speech. Whether this discrepancy stems from our adaptations not working well for Whisper Small, the limited informational content of the data, or another factor remains an open question.

Model	Regular Speech	Tali's *In-Domain* Speech
Tiny Baseline	0.447	0.452
Tiny Adapted	0.345	0.402
Base Baseline	0.303	0.361
Base Adapted	0.312	0.349
Small Adapted	0.209	0.319

Table 7.8.: Word Error Rates (WER) for regular and Tali's speech in the *In-Domain* setting.

To sum up, we find it highly likely that in order to work well, Whisper will have to learn from the data collected during the deployment. Fortunately, the results for artificial conversations give us hope that when provided with enough data from the target domain, Whisper will achieve good performance.

8. Deployment

In this chapter, we discuss deploying the model in the real world. We experiment with several transcription methods and evaluate their practical usefulness. Finally, we propose an application deploying our model.

8.1. Direct Transcription

Although Whisper [1] comes with pre-implemented methods for speech transcription, these do not perform well in Tali's case. Specifically, they tend to hallucinate words, likely due to her slow rate of speech and the presence of long pauses. To clean the audio and mitigate these issues, we asked Tali to make deliberate pauses between her words for several recording sessions (those were not used for training our model). This practice is recommended by her therapists and should, by itself, improve the intelligibility of her speech. Additionally, it enabled us to isolate only the speech-relevant segments of audio by extracting the N loudest segments, where N is the number of words in the corresponding line of the *artificial conversation* script. These segments were then concatenated and separated by 500 ms of silence, simulating realistic pauses between words. When these resulting audio segments were passed to Whisper for direct transcription, we obtained the results presented in Table 8.1.

True transcription	**Predicted transcription**
Chápu naučím vás zemské etiketě	chápu naučím nás lidské endemie
To jsou pravidla slušnosti	To jsou pravidla a slušností
Začněme s podáním ruky	Začneme podáním luky
Stoupnete si usmějete se	stoupnit. Usmějte se.
Ruku podáváte ze předu	luku po tvrdém řadu předu.

Table 8.1.: Comparison of true and predicted transcriptions with word-level colouring: the green words perfectly match the ground truth and the orange are close to it.

8.2. Word-by-Word Transcription

We find the transcriptions produced by the method above reasonable, and so do Tali and her family (see their feedback in Figure 8.1). However, in order to provide users with intermediate responses (i.e., partial transcriptions before the full sentence is completed) and to alleviate the impact of high inaccuracy, we further experiment with word-by-word transcription. This approach also allows us to present multiple candidate transcriptions to the user, enhancing the system's responsiveness and transparency.

As mentioned in the previous section, we tested our deployment strategies on *artificial conversations*, where Tali was instructed to make at least one-second pauses between her words. This enabled us to automatically split the recorded conversational lines into word-level segments, as detailed above.

After splitting each of the conversational lines by Tali to shorter segments containing exactly one word, we compute the top five transcriptions for each word and measure the relative frequency of correct word-level transcriptions.

We experiment with the following strategies, using *Small Adapted* model and a beam search of width 15 in all cases[1]. The performance of each strategy is presented in Table 8.2. We provide the implementation of the transcription strategies in the Appendix in Section D.

- **Unguided beam search transcription**: We apply standard beam search to obtain the five most likely word-level transcriptions. Each word segment is transcribed independently of the others, and we constrain the output to not exceed one word. Despite being provided very limited context, this strategy achieves results comparable to those of more advanced transcription approaches.
- **Guided beam search transcription**: We again apply standard beam search, but constrain the possible outputs to existing Czech words (as provided by Github repository *All Words in All Languages*[2]) stored in a prefix tree [53]. When predicting a new token, we mask all logits except for those representing valid suffixes according

[1] At each step of beam search, we track the N most likely token sequences. We then extend each of these sequences by predicting the probabilities of the next token, and from all the possible extensions, we again select the N with the highest overall probabilities to continue the search.

[2] https://github.com/eymenefealtun/all-words-in-all-languages

to the prefix tree. Each word-level segment is transcribed independently from the others.

- **Guided beam search transcription with merged segments**: We apply the same guided beam search with the Czech vocabulary prefix tree, but additionally merge each word segment with its preceding segments and provide the model with the true transcriptions of the previous words. While this method yields the best results, its deployment would likely require real-time, accurate feedback from Tali or her conversation partner so that the model has access to true transcriptions of previous segments. This may only be feasible to a limited extent.

These strategies lead to the results presented in Table 8.2. The time per word was measured on a CPU-only device, indicating that stronger hardware would be necessary for a real-world application.

Strategy	Top-1 Accuracy	Top-2 Accuracy	Top-5 Accuracy	Time per word (s)
Unguided	0.29	0.35	0.42	5.5
Guided	0.30	0.34	0.40	**4.8**
Guided + Merged	**0.36**	**0.41**	**0.46**	4.9

Table 8.2.: Word-level accuracies for different decoding strategies

8.3. Feedback from Tali

Tali and her family confirmed to us that the results achieved by direct transcription with dropped silent segments and *guided beam search transcription with merged segment* would already be helpful to them (especially for communication with Tali's assistant in school and with her therapist). Overall, they provided us with the following feedback (translated from Czech to English):

> It looks pretty good from the preview. Tali has quite learned how to separate words, so she keeps to it when speaking, which helps everything. Another side effect of the recording is that she strengthens her voice by recording. Even to the point where it's noticeable. So overall, our efforts are beneficial on many levels.

Figure 8.1.: Feedback from Tali's mother (we received also several others, but this is the most elaborate one)

8.4. Application Architecture

After receiving such positive feedback, we shift our focus to the development of an application that would enable the use of the system in real-world scenarios. While developing an application is planned as future work and thus lies beyond the scope of this thesis, we provide a high-level outline of the envisioned system for the interested reader:

1. The system shall be implemented as a web application accessible through a web browser.
2. When Tali opens the application, her phone's microphone shall be activated, and a link for her conversation partners shall be provided in the form of a QR code.
3. Conversation partners can connect to the application using the link generated by Tali's application.
4. An algorithm running in the background shall sample segments of Tali's speech from the audio stream. The exact nature of this mechanism is yet to be determined. Currently, we plan to use voice embeddings from pyannote-audio [46] to decide whether the incoming audio segments contain Tali's speech or not, based on similarity measures.
5. Another computationally powerful device shall connect to the web server as a *transcribing client*.
6. Segments identified as containing Tali's speech shall be sent to the server, which will forward them to the *transcribing client*.

7. The *transcribing client* shall return the most likely transcriptions, and the web application shall display them to both Tali and her conversation partners.
8. Conversation partners may select the correct transcription. If they do so, the ASR system shall use it as contextual input for transcribing future segments, and it shall save it for further training.
9. Alternatively, conversation partners can indicate that the provided transcriptions were incorrect and that Tali should repeat the respective word.
10. The feedback from the points above shall be used to further improve the system through continued training.
11. The web application shall also provide an interface for creating purpose-specific vocabularies to enhance performance in specialised conversations. However, the exact method for constructing these vocabularies remains an open problem.

For better intuition we also provide a diagram of the proposed application in Figure 8.2.

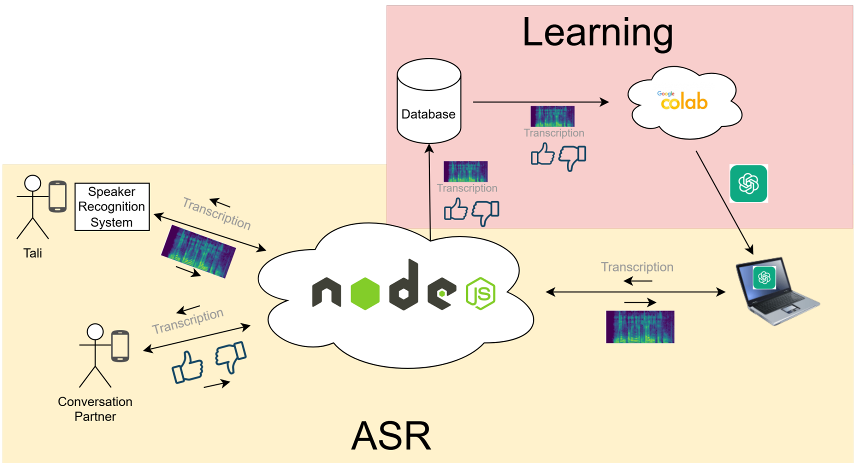

Figure 8.2.: Scheme of the proposed application.

This design is not optimal for long-term usage as it requires a transcribing client (initially our laptop) to connect to the web server each time the application is used. However, it can be easily developed with the resources currently available, making it well-suited for rapid and low-cost prototype deployment. Furthermore, upgrading to a more powerful client should be straightforward and would only require launching the transcribing client on a new device.

9. Conclusion

This thesis investigates the development of an assistive automatic speech recognition system for a patient with a severe speech impediment caused by a tracheostomy and possibly also by neurological damage.

We collect and annotate twenty four hours of her voice. To this end, we develop a novel data collection method: artificial conversations. This strategy greatly reduces the time required for the data labelling. We analyze different methods of data collection. We also create a public dataset of the Tali's speech. Interstingly, the feedback from Tali's family indicates that the artificial conversations have also positive logopedic effects.

We train Whisper models of various sizes on the collected data, utilizing pre-training on both regular Czech speech and quasi-tracheostomy speech. Additionally, we incorporate knowledge distillation (KD) from a masked language model (MLM) into Whisper. These adaptations were initially selected based on experiments with Whisper Tiny and resulted in substantial improvements for this model. In particular, KD from the MLM led to significant gains even for regular Czech speech. However, for larger models, these adaptations appear to provide diminishing returns.

As our models are not yet sufficiently accurate for standard end-to-end transcription, we developed several strategies for real-world decoding. These approaches achieve performance that substantially surpasses that of untrained human listeners and are regarded as helpful by both the patient and her family. Finally, we outline a plan for an application designed to make this system accessible and supportive for the patient.

9.1. Future work

As already mentioned, although our system is usable, it still does not reach the accuracy of Tali's family members or caretakers, indicating that there remains room for improvement. Specifically, we believe that creating a more realistic quasi-tracheostomy voice model or ensembling Whisper with lip-reading models and large language models (LLMs) for enhanced semantic understanding are promising directions for future work.

At the same time, the system would benefit from reduced computational expenses. Further improving Whisper Tiny performance combined with efficiency measures such as quantization might remove the need for an external server thus cutting the financial burden of our application dramatically.

Nevertheless, as our tool performs sufficiently well even in its current form, we find it reasonable to put it into practical use as soon as possible and let the real-world insights guide our research.

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Appendix

A. Prompt for Generating Artificial Conversations

In the code block below, we provide the prompt that we used to generate artificial conversations.

Variables inserted into this prompt are:

- *content_file*: A content of file providing additional material for the conversation (e.g. a copy of a related Wikipedia article).
- *plot*: A brief summary of what the conversation should be about.
- *prompt_dialogue*: A previous dialogue to be continued on.

```
f"""Consider the following text:

{content_file}

Continue the provided dialogue of Tomka and Šalina (no other character)
by at least 70 more lines where: {plot} Tomka uses at most five words
per sentence. Šalina at most five. Tomka is a female director and CEO of
space company ZetorSpace.
Keep the vocabulary simple. Take care of the correct Czech word order.

Make sure to stick to the line format: '<speaker_name>:<content>'

{prompt_dialogue}"""
```

We generate a conversation script by prompting the LLM several times. Each time we use a different *plot* and with the *prompt_dialogue* extended by previously generated scripts.

B. Artificial Conversation Script

The conversation below was used to test the prototype of our application. Collection of training data was only a secondary objective. The length and the topic of the conversation correspond to that.

B.1. Czech Original

```
Tomka: Ahoj, Šalino
Šalina: Ahoj, Tomko.
Tomka: Co to zrovna bouchlo?
Šalina: Asi nová raketa.
Tomka: To bych se nedivila.
Šalina: Ty vybuchují často.
Tomka: Nové modely často selhávají.
Šalina: Je to nepříjemné, ale efektivní.
Tomka: A letos opakované.
Šalina: Jaké konkrétní problémy jste řešili?
Tomka: Motory se neodpálily.
Šalina: To je velký problém.
Tomka: Selhaly kvůli softwaru.
Šalina: Musíme se učit z chyb.
Tomka: Přesně tak.
Šalina: Jaké další oblasti to platí?
Tomka: Ve výrobě automobilů.
Šalina: Tam je také hodně pokusů.
Tomka: A v medicíně je to stejné.
Šalina: Jaké chyby se tam dělají?
Tomka: Léky mohou mít vedlejší účinky.
Šalina: To je velmi vážné.
```

Tomka: Ale zlepšuje to další léky.

Šalina: Učení se z chyb je důležité.

Tomka: Ano, i ve vzdělávání to platí.

Šalina: Jak to myslíš?

Tomka: Děti se učí chyby opravovat.

Šalina: To je důležitý proces.

Tomka: Raketový výzkum je podobný.

Šalina: Analyzujeme každý neúspěch důkladně.

Tomka: Poté vylepšujeme design raket.

Šalina: Jak rychle to obvykle jde?

Tomka: Hodně záleží na problému.

Šalina: Někdy trvá měsíce, že?

Tomka: Někdy dokonce roky.

Šalina: A co ostatní společnosti?

Tomka: Také zažívají podobné situace.

Šalina: A co zákazníci?

Tomka: Ti musí mít trpělivost.

Šalina: To není snadné, že?

Tomka: Není, ale je to důležité.

Šalina: Jak zákazníky informujete?

Tomka: Pravidelně je aktualizujeme.

Šalina: Máte nějaké speciální kanály?

Tomka: Ano, používáme sociální sítě.

Šalina: To je skvělý způsob komunikace.

Tomka: A videa prezentují pokrok.

Šalina: Jaké další metody využíváte?

Tomka: Webináře a online konference.

Šalina: Kolik máte zaměstnanců?

Tomka: Přes padesát.

Šalina: Je to dostatečné?

Tomka: Zatím ano, rozrůstáme se.

Šalina: Jaké jsou vaše plány?

Tomka: Vyvinout nový typ rakety.

Šalina: A to chce hodně práce, že?

Tomka: Ano, ale my to zvládneme.
Šalina: Učení se z chyb pomůže.
Tomka: Přesně. Víra v tým je klíčová.
Šalina: Bez podpory to nejde.
Tomka: Všichni se snaží odvést své maximum.
Šalina: A to vytváří skvělou atmosféru.
Tomka: Spolupráce je náš motor pokroku.
Šalina: Jak jste na tom s financemi?
Tomka: Pracujeme na získání investic.
Šalina: Jaké investory hledáte?
Tomka: Technologické a inovativní společnosti.
Šalina: A co fondy a granty?
Tomka: Také se snažíme o granty.
Šalina: Co vaše konkurence?
Tomka: Ta je silná, ale my se nebojíme.
Šalina: Výzkum a vývoj jsou základ.
Tomka: Ano, bez nich se neposuneme.
Šalina: Tím se prohlubuje znalost.
Tomka: Přesně tak, z chyb se učí.
Šalina: To je udržitelné pro budoucnost.
Tomka: A naše mise zůstává stejná.
Šalina: Jaká je vaše mise?
Tomka: Vytvářet bezpečné a efektivní rakety.
Šalina: Dobře, téma je fascinující.
Tomka: Děkuji, miluji svoji práci.

B.2. English Translation

Tomka: Hi, Shalina.
Shalina: Hi, Tomka.
Tomka: What just exploded?
Shalina: Probably a new rocket.
Tomka: I wouldn't be surprised.
Shalina: They explode often.

Tomka: New models fail often.
Shalina: It’s unpleasant, but effective.
Tomka: And it’s happened repeatedly this year.
Shalina: What specific problems have you dealt with?
Tomka: The engines didn’t launch.
Shalina: That’s a big problem.
Tomka: They failed due to the software.
Shalina: We need to learn from our mistakes.
Tomka: Exactly.
Shalina: What other areas does this apply to?
Tomka: In car manufacturing.
Shalina: There are also many trials there.
Tomka: And it’s the same in medicine.
Shalina: What mistakes are made there?
Tomka: Medications can have side effects.
Shalina: That’s very serious.
Tomka: But it improves other medications.
Shalina: Learning from mistakes is important.
Tomka: Yes, it applies to education too.
Shalina: What do you mean by that?
Tomka: Children learn to correct mistakes.
Shalina: That’s an important process.
Tomka: Rocket research is similar.
Shalina: We analyze every failure thoroughly.
Tomka: Then we improve the rocket designs.
Shalina: How quickly does that usually go?
Tomka: It depends a lot on the problem.
Shalina: Sometimes it takes months, right?
Tomka: Sometimes even years.
Shalina: And what about other companies?
Tomka: They also experience similar situations.
Shalina: And what about the customers?
Tomka: They need to be patient.
Shalina: That’s not easy, is it?

Tomka: It's not, but it's important.
Shalina: How do you keep customers informed?
Tomka: We update them regularly.
Shalina: Do you have any special channels?
Tomka: Yes, we use social media.
Shalina: That's a great way to communicate.
Tomka: And videos showcase progress.
Shalina: What other methods do you use?
Tomka: Webinars and online conferences.
Shalina: How many employees do you have?
Tomka: Over fifty.
Shalina: Is that sufficient?
Tomka: For now, yes, we're growing.
Shalina: What are your plans?
Tomka: To develop a new type of rocket.
Shalina: And that requires a lot of work, right?
Tomka: Yes, but we can handle it.
Shalina: Learning from mistakes will help.
Tomka: Exactly. Belief in the team is key.
Shalina: It doesn't work without support.
Tomka: Everyone is trying to do their best.
Shalina: And that creates a great atmosphere.
Tomka: Collaboration is our engine for progress.
Shalina: How are you doing financially?
Tomka: We're working on securing investments.
Shalina: What kind of investors are you looking for?
Tomka: Technology and innovative companies.
Shalina: What about funds and grants?
Tomka: We're also trying for grants.
Shalina: What about your competition?
Tomka: They are strong, but we are not afraid.
Shalina: Research and development are fundamental.
Tomka: Yes, without them we won't progress.
Shalina: That deepens knowledge.

```
Tomka: Exactly, we learn from mistakes.
Shalina: That's sustainable for the future.
Tomka: And our mission remains the same.
Shalina: What is your mission?
Tomka: To create safe and efficient rockets.
Shalina: Well, the topic is fascinating.
Tomka: Thank you, I love my job.
```

C. Script for Removing the Periodic Source of the Speech

```
def apply_lpc_noise_resynthesis(samples, radLPC=12) -> AudioSegment:
    # Remove DC component
    samples -= np.mean(samples)

    # Pre-emphasis
    pre_emphasis = 0.8
    samples = scipy.signal.lfilter([1, -pre_emphasis], 1, samples)

    # Parameters for segmentation
    window_length = 512
    shift_ratio = 0.5
    shift = int(window_length * shift_ratio)

    # Hamming window and scaling
    window = np.hamming(window_length + 1)[:-1]
    constant = np.sum(window) / window_length / shift_ratio
    window /= constant

    # Output and coefficient placeholders
    output = np.zeros(len(samples))
    num_segments = (len(samples) - shift) // shift
    coefficients = np.full((num_segments, radLPC), np.nan)
```

```
    segment_index = 0
    for i1 in range(0, len(samples) - window_length + 1, shift):
        i2 = i1 + window_length
        segment = samples[i1:i2] * window

        if np.var(segment) > 0:
            correlated_segments = np.correlate(segment, segment, mode='full')
            r = correlated_segments[window_length-1:window_length+radLPC]
            try:
                ar_coeffs = scipy.linalg.solve_toeplitz((r[:-1], r[:-1]), r[1:])
                coefficients[segment_index, :len(ar_coeffs)] = ar_coeffs

                excitation = np.random.uniform(-1, 1, window_length)
                estimated_segment = scipy.signal.lfilter(
                    [np.sqrt(np.var(segment))],
                    np.concatenate([[1], -ar_coeffs]),
                    excitation
                )

                output[i1:i2] += estimated_segment
            except np.linalg.LinAlgError:
                pass  # Skip segments that can't be solved due to numerical issues

        segment_index += 1

    # De-emphasis
    output = scipy.signal.lfilter([1], [1, -pre_emphasis], output)

    # Normalize
    output /= np.max(np.abs(output))

    return output
```

D. Word-by-Word Transcription Strategies

All our word-by-word decoding strategies are facilitated by the following function:

```
def decode_beam(self, audio_segment, beam_size=5,
    n_results = 3, prev_trans = "", guided = True):
    tokenize = lambda t: self.tokenizer.encode(t, allowed_special="all")
    assert beam_size>=n_results,"N results to be returned cannot exceed the beam size"
    with torch.no_grad():
        # Prepare input
        audio = whisper.pad_or_trim(audio_segment)
        mel = whisper.log_mel_spectrogram(audio, n_mels=80)
        encoder_output = self.model.encoder(mel.unsqueeze(0))

        # Initialize search
        trans_start = "<|startoftranscript|><|cs|><|transcribe|><|notimestamps|>"
        initial_tokens = tokenize(f"{trans_start}{prev_trans}")
        token_dict = self.next_token_dict if prev_trans else self.first_token_dict
        beam = [(initial_tokens, 0.0, token_dict)]

        # Beam search loop
        for e in range(25):  # Max length of sequence
            new_beam = []
            # Reove finished hypotheses
            removed_hypotheses = []
            for b, el_tuple in enumerate(beam):
                tokens, _, _ = el_tuple
                if tokens[-1] == self.tokenizer.eot:
                    new_beam.append(el_tuple)
                    removed_hypotheses.append(b)
            for h_idx in removed_hypotheses[::-1]: del beam[h_idx]
            if not len(beam):
                beam = new_beam
                break  # Stop early if all hypotheses are finished
```

```python
# Prepare batched inputs
token_batch = [tokens for tokens, _, _ in beam]
current_dicts = [current_dict for _, _, current_dict in beam]
long_batch = torch.tensor(token_batch).long()
encoded_txt_tokens = long_batch.to(encoder_output.device)

# Decode batch
logits = self.model.decoder(encoded_txt_tokens, encoder_output)[:, -1, :]

if guided:
    # Apply individual masks for each beam
    vocab_size = logits.shape[-1]
    mask_shape = (len(beam), vocab_size)
    masks = torch.full(mask_shape, float('-inf'), device=logits.device)

    for i, current_dict in enumerate(current_dicts):
        allowed_indices = list(current_dict.keys())
        masks[i, allowed_indices] = 0  # Allow only valid tokens

    logits = logits + masks  # Mask invalid tokens

log_probs = torch.log_softmax(logits, dim=-1)  # Compute log probabilities

# Get top-k candidates for each beam hypothesis
topk_log_probs, topk_indices = torch.topk(log_probs, beam_size, dim=-1)

# Expand beam
for i in range(len(beam)):
    for j in range(beam_size):
        new_token = topk_indices[i, j].item()
        new_log_prob = beam[i][1] + topk_log_probs[i, j].item()
        new_dict = beam[i][2].get(new_token, {})  # Navigate token tree
        new_beam.append((beam[i][0] + [new_token], new_log_prob, new_dict))
```

```
            # Keep top beam_size hypotheses
            beam = sorted(new_beam, key=lambda x: x[1], reverse=True)[:beam_size]

        # Return best sequence
        best_tokens = [(beam_el[0],beam_el[1]) for beam_el in beam[:n_results+1]]
        # Put together the results
        result = []
        for tokens,log_prob in best_tokens:
            decoded_word = ""
            if self.tokenizer.decode(tokens[len(initial_tokens):-1]).split():
                whole_text = self.tokenizer.decode(tokens[len(initial_tokens):-1])
                decoded_word = whole_text.split()[0]
            else: decoded_word = ’’
            result.append((decoded_word,log_prob))
        return result
```

The implementations of individual decoding strategies are then detialed in the following subsections.

D.1. Unguided Beam Search Transcription

```
def unguided_transcribe(chunk, merged_segment,prev_trans):
    return  decoder.decode_beam(chunk, 15,5,guided=False)
```

D.2. Guided Beam Search Transcription

```
def guided_transcribe(chunk, merged_segment,prev_trans):
    return decoder.decode_beam(chunk, 15,5,guided=True)
```

D.3. Guided Beam Search Transcription with Merged Segments

```
def guided_and_merging_transcribe(chunk, merged_segment, prev_trans):
    decoder.decode_beam(merged_segment, 15,5,guided=True,
    previous_transcription=prev_trans)
```